INFO 241: Problem Set 2

Find this and Change it to your Own Name

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Dedication Page

This is the second problem set. I dedicate it to [fill in with your dedication.] :joy:

What happens when pilgrims attend the Hajj pilgrimage to Mecca?

What happens when a diverse set of people are brought together toward a common purpose? Perhaps it brings diverse sets of people together in religious community who learn about each others' commonalities. Or, perhaps, the event highlights the differences between groups.

If one were to simply speak with people who *choose* to go on a pilgrimage and compare views to those who *choose* not to go on that pilgrimage, there are **obvious** issues of selection bias. People who choose to go may be more open to new experiences, or more religious, or more conservative or … when compared to people who choose not to go.

In order to test the causal effect of attending the Hajj on views about others, you would need to conduct a randomized experiment. Clingingsmith, Khwaja and Kremer (2009) did just that!

In 2006, Pakistan conducted a lottery to allocate visas to attend the Hajj. The researchers conduct surveys among people who applied to this visa lottery, and talk to two different groups of people:

1. One group is a set of people who applied for the Hajj visa and were successful;
2. The other group is a set of people who applied for the Hajj visa and were not successful.

Because this was a randomly assigned lottery, these two groups of people are randomly assigned, and so meet our requirements of a valid experiment!

The random assignment and data is collected in the following way (detailed in the paper):

- Pakistani nationals apply for a chance to attend the Hajj at a domestic bank. During the time period of the study, 2006, Saudi officials agreed to grant 150,000 visas.
- Of the 135,000 people who applied for a visa, 59% of those who applied were successful.
- The remainder of the visas were granted under a different allocation method that was not randomly assigned, and so this experiment cannot provide causal evidence about other allocation mechanisms.
- Among the 135,000 who apply, the authors conduct a random sample and survey about these respondents views about others.

1 Reading the Paper

1.1 Interview Timing

When do the researchers interview Pakistani's about their views? Before the lottery? After the lottery but before the Hajj? After the Hajj?

[Fill this in.]

1.2 Interview Mode

How are the interviews conducted? Over the phone? Over email? In person?

[Fill this in.]

1.3 What is AES?

On page 1144 the researchers describe a crucial term that they reuse repeatedly throughout the paper, the AES. What does this term stand for?

[Fill this in.]

1.4 AES Formula

The researchers go on to define the AES in the following way:

"For a family of J related outcomes Y^j in an index, with Hajj local average treatment effects of π_j, the AES is

$$\tau = \frac{1}{J} \sum_{j=1}^{J} \frac{\pi_j}{\sigma_j},$$

where σ_j is the standard deviation of outcome j in the comparison group. Why does this feel like a weighted average? What is the quantity that is being used to do the weighting?

[Fill this in.]

1.5 Practical advice

I've (Alex…) has worked through this data quite a bit. It turns out that the weights are not really that important either to the statistical results or their interpretations.

Throughout the rest of the analysis, when you're asked about a concept, you can do the following: Average all of the scores for that index into a single value and conduct your analysis against this average.

I've asked you to think through the AES, and then simplified it so that we can use the data in a more straightforward way to practice with our randomization inference tools.

2 Now, Analyze the Hajj Data!

```
library(data.table)
library(here)
```

```
here() starts at /Users/douglashughes/iSchool/experiments/homework/problem-set-2-starter
```

Now that you have thought about the design of the experiment, conduct analysis that is consistent with the paper. The researchers write that they use the AES, which you have reasoned about in an earlier question. I've given advice that instead of using this more complex concept, instead we will simply create the averages before the analysis.

2.1 State a null hypothesis

State the sharp-null hypothesis that you will be testing.

[Fill this in.]

2.2 Load the data

Load the data and store it in a variable called `d`.

```
d <- fread(here("data", "processed", "hajj_public.csv"))
```

2.3 Create the average views

There are several variables that contain views about people from other areas. These are stored in the variables that start with `views_`. Create a new variable called `views_others` that is the average of these variables.

Create a plot of the views toward others that you this is informative about this data. It could be a histogram, a boxplot, or whatever form you this is most informative to a reader. It would be a bonus if you could also show the views toward individual groups, but the main point is to communicate information about `views_others`.

What, if anything do you note about this data?

[Fill this in.]

2.4 Group by average

Using `data.table`, group the data by `won_lottery` and report whether views toward others are generally more positive among lottery winners or lottery non-winners. This answer should be of the form `d[ , .(mean_views = ...), keyby = ...]` where you have filled in the `...` with the appropriate functions and variables.

```
# the result should be a data.table with two columns and two rows
hajj_group_mean <- 'fill this in'

# from the `hajj_group_mean` produce a single, numeric vector that is the ate.
# check that it is numeric using `class(hajj_ate)`
hajj_ate        <- 'fill this in'
```

How much do views about other differ between those who won the lottery and those who did not?

[Fill this in.]

2.5 Randomization inference: Simulate the sharp null hypothesis

But is this a "meaningful" difference? Or, could a difference of this size have arisen from an "unlucky" randomization? Conduct 10,000 simulated random assignments under the sharp null hypothesis to find out. (Do not just copy the code from the async, think about how to write this yourself.)

```
## do your work to conduct the randomization inference here.
## as a reminder, RI will randomly permute / assign the treatment variable
## and recompute the test-statistic (i.e. the mean difference) under each permutation
## this should be a numeric vector that has a length equal to the number
## of RI permutations you ran
hajj_ri_distribution <- 'fill this in'
```

Create a plot of `hajj_ri_distribution` that is informative about the distribution of estimated treatment effects *if the sharp null hypothesis were true.*

2.6 Randomization inference: At least as large

How many of the simulated random assignments generate an estimated ATE that is at least as large as the actual estimate of the ATE?

Conduct your work in the code chunk below, saving the results into `hajj_count_larger`, but also support your coding with a narrative description. In that narrative description (and throughout), use R's "inline code chunks" to write your answer consistent with each time your run your code.

```
# length 1 numeric vector from comparison of `hajj_ate` and `hajj_ri_distribution`
hajj_count_larger <- 'fill this in'
```

[Fill this in.]

2.7 Randomization inference: one-sided p-value

If there are `hajj_count_larger` (fill this in) randomizations that are larger than `hajj_ate` (fill this in), what is the *one-tailed* p-value? Both write the code in the following chunk, and include a narrative description of the result following your code.

```
# length 1 numeric vector
hajj_one_tailed_p_value <- 'fill this in'
```

[Fill this in.]

2.8 Randomization inference: two-sided p-value

Now, conduct a similar test, but for a two-sided p-value. You can either use two tests, one for larger than and another for smaller than; or, you can use an absolute value (`abs`). Both write the code in the following chunk, and include a narrative description of the result following your code.

```
# length 1 numeric vector
hajj_two_tailed_p_value <- 'fill this in'
```

[Fill this in.]

3 Sports Cards

```
library(data.table)
library(here)
```

In this experiment, the experimenters invited consumers at a sports card trading show to bid against one other bidder for a pair trading cards. We abstract from the multi-unit-auction details here, and simply state that the treatment auction format was theoretically predicted to produce lower bids than the control auction format. We provide you a relevant subset of data from the experiment.

In this question, we are asking you to produce p-values and confidence intervals in three different ways:

1. Using a `t.test`;
2. Using randomization inference; and,
3. Using a regression with robust standard errors.

```
d <- fread(here("data", "processed", "list_data.csv"))
```

3.1 t-test and confidence interval

Using a `t.test`, compute a 95% confidence interval for the difference between the treatment mean and the control mean. After you conduct your test, write a narrative statement, *using inline code evaluation* that describes what your tests find, and how you interpret these results. (You should be able to look into `str(t_test_cards)` to find the pieces that you want to pull to include in your written results.)

```
# this should be the t.test object. Extract pieces from this object
# in-text below the code chunk.
t_test_cards <- 'fill this in'
```

[Fill this in.]

3.2 Interpretation of confidence interval

In your own words, what does this confidence interval mean? This can be simple language, but it has to be statistically appropriate language.

[Fill this in.]

3.3 Randomization inference, and confidence interval?

Conduct a randomization inference process, with `n_ri_loops = 1000`, using an estimator that you write by hand (i.e. in the same way as earlier questions). On the sharp-null distribution that this process creates, compute the 2.5% quantile and the 97.5% quantile using the function `quantile` with the appropriate vector passed to the `probs` argument. This is the randomization-based uncertainty that is generated by your design. After you conduct your test, write a narrative statement of your test results.

```
## first, do you work for the randomization inference
n_ri_loops <- 1000

cards_ate             <- 'fill this in'
cards_ri_distribution <- 'fill this in' # numeric vector of length equal
                                        # to your number of RI permutations
cards_ri_quantiles    <- 'fill this in' # there's a built-in to pull these.
cards_ri_p_value      <- 'fill this in'
```

[Fill this in.]

3.4 Compare regression and randomization inference

Do you learn anything different if you regress the outcome on a binary treatment variable? To answer this question, regress `bid` on a binary variable equal to 0 for the control auction and 1 for the treatment auction and then calculate the 95% confidence interval using *classical standard errors* (in a moment you will calculate with *robust standard errors*). There are two ways to do this – you can code them by hand; or use a built-in, `confint`. After you conduct your test, write a narrative statement of your test results.

```
# this should be a model object, class = 'lm'.
mod <- 'fill this in'
```

[Fill this in.]

3.5 Regression with robust confidence interval

Calculate the 95% confidence interval using robust standard errors, using the `sandwich` package. There is a function in `lmtest` called `coefci` that can help with this. It is also possible to do this work by hand. After you conduct your test, write a narrative statement of your test results.

```
# this should be a numeric vector of length 2
cards_robust_ci <- 'fill this in'
```

[Fill this in.]

3.6 Compare and contrast results

Characterize what you learn from each of these different methods – are the results contingent on the method of analysis that you choose?

[Fill this in.]

4 Power Analysis

Understanding whether your experiment design and data collection strategy are able to reject the null hypothesis *when they should* is valuable! And, this is not a theoretical value. If your design and data collection cannot reject the null hypothesis, why even run the experiment in the first place?

The classical formulation of power asks, "Given a test procedure and data, what proportion of the tests I *could conduct* would reject the null hypothesis?"

Imagine that you and David Reiley are going to revive the sports card experiment from the previous question. However, because it is for a class project, and because you have already spent all your money on a shiny new data science degree :raised_hands: :money_with_wings: , you are not going to be able to afford to recruit as many participants as before.

4.1 Describe your testing procedure

Describe a t-test based testing procedure that you might conduct for this experiment. What is your null hypothesis, and what would it take for you to reject this null hypothesis? (This second statement could either be in terms of p-values, or critical values.)

[Fill this in.]

4.2 Suppose you only had ten subjects, what would you learn

Suppose that you are only able to recruit 10 people to be a part of your experiment – 5 in treatment and another 5 in control. Simulate "re-conducting" the sports card experiment once by sampling from the data you previously collected, and conducting the test that you've written down in part 1 above. *Given the results of this 10 person simulation, would your test reject the null hypothesis?*

```
# this should be a test object
t_test_ten_people <- 'fill this in'
```

[Fill this in.]

4.3 With only ten subjects, what is your power?

Repeat this process – sampling 10 people from your existing data and conducting the appropriate test – one-thousand times. Each time that you conduct this sample and test, pull the p-value from your t-test and store it in an object for later use. *Consider whether your sampling process should sample with or without replacement.*

```
# fill this in with the p-values from your power analysis
t_test_p_values <- rep(NA, 1000)

## you can either write a for loop, use an apply method, or use replicate
## (which is an easy-of-use wrapper to an apply method)
```

[Fill this in.]

4.4 Visual analysis

Use `ggplot` and either `geom_hist()` or `geom_density()` to produce a distribution of your p-values, and describe what you see. *What impression does this leave you with about the power of your test?*

[Fill this in.]

4.5 Interpret your results, given your power

Suppose that you and David were to actually run this experiment and design – sample 10 people, conduct a t-test, and draw a conclusion. **And** suppose that when you get the data back, **lo and behold** it happens to reject the null hypothesis. Given the power that your design possesses, does the result seem reliable? Or, does it seem like it might be a false-positive result?

[Fill this in.]

4.6 Conduct a power analysis

F. Apply the decision rule that you wrote down in part 1 above to each of the simulations you have conducted. What proportion of your simulations have rejected your null hypothesis? This is the p-value that this design and testing procedure generates. After you write and execute your code, include a narrative sentence or two about what you see.

```
t_test_rejects <- 'fill this in'
```

[Fill this in.]

4.7 Moar power!

Does buying more sample increase the power of your test? Apply the algorithm you have just written onto different sizes of data. Namely, conduct the exact same process that you have for 10 people, but now conduct the process for every 10% of recruitment size of the original data: Conduct a power analysis with a 10%, 20%, 30%, … 200% sample of the original data. (You could be more granular if you like, perhaps running this task for every 1% of the data).

```
percentages_to_sample <- 'fill this in'
```

[Fill this in.]

5 Solution Set

There's nothing to go here. But, once Alex has made it available to the class, the solution set will be available at the following link: [link].